NEXUS

Non-Equilibrium eXploration with Unified Synthesis

A dependency-free drug discovery pipeline, with measured results

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What this is

NEXUS is a drug discovery pipeline whose entire scientific stack runs on NumPy alone: SMILES parsing, molecular graphs, fingerprints, descriptors, 3D conformers, automatic differentiation, message-passing neural networks, graph diffusion models, Gaussian processes, persistent homology, discrete Ricci curvature, conformal prediction, non-equilibrium free energy estimators and the Gene Ontology. There is no RDKit and no PyTorch.

Every component is checked against something external — published reference values, exactly solvable physics, known topology, or a falsifiable statistical guarantee. This document is generated from the files in `results/`, so the numbers in it are the measured ones. Results that did not help are reported alongside those that did.

Python in the package	7,119 lines
Tests	1,823 lines
Benchmarks and validation	2,156 lines
Mantis panel (TypeScript)	804 lines
Total Python	11,098 lines
Test suite	268 tests, all passing

1 Validation summary

Component	Check	Result
SMILES parser	parse the whole MoleculeNet suite	59,974 of 59,975 molecules (99.998%)
Descriptors	against reference columns in ESOL and BACE	MW, heavy atoms, rings exact; TPSA $r = 0.9998$
TPSA	published values for known drugs	aspirin 63.6, caffeine 58.44, ibuprofen 37.3 — exact
Scaffold split	leakage into the held-out set	0.000, against 35–78% for random splits
Autograd	every operation vs. finite differences	42 operations, all agree
Conformers	bond lengths and ring planarity	error < 0.05 Å; aromatic rings planar
Persistent homology	shapes with known topology	circle $\to$ 1 loop; 2 clusters $\to$ Betti-0 $= 2$
Ricci curvature	optimal transport vs. hand-computed optima	exact on all six; $\kappa = 0$ on cycles
Conformal prediction	empirical vs. nominal coverage	matches across α, models, 200 trials
Free energy	harmonic oscillator, exact $\Delta F = 0.6931$	BAR 0.6957, Crooks 0.6899, Jarzynski 0.6979
Active learning	retrospective screen of 8,000 compounds	3.16× enrichment (random control 1.01×)
Gene Ontology	enrichment and semantic similarity	4/5 controls at rank 1; similarity AUC 0.885

2 The pipeline running

The screenshots below are of the real panel driving the real service: the Python package computes every number shown, and the panel reads them over HTTP. Nothing is mocked except the four Mantis store modules that supply the selected points.

3 Property prediction

Scaffold-balanced splits, three seeds, early stopping on validation, one evaluation of the selected model on the held-out test set.

dataset	n	metric	NEXUS D-MPNN	published[1]	best	best method
bace	1513	roc_auc	0.853 ± 0.015	0.857	0.906	ECFP+descriptors + RF
bbbp	2050	roc_auc	0.911 ± 0.006	0.913	0.934	ECFP + RF
clintox	1484	roc_auc	0.896 ± 0.041	0.897	0.897	D-MPNN + descriptors

[1] Yang et al. (2019), same protocol but not identical splits — shown for orientation, not as a like-for-like comparison.

The honest reading. The D-MPNN reproduces the published numbers closely, but tuned fingerprint

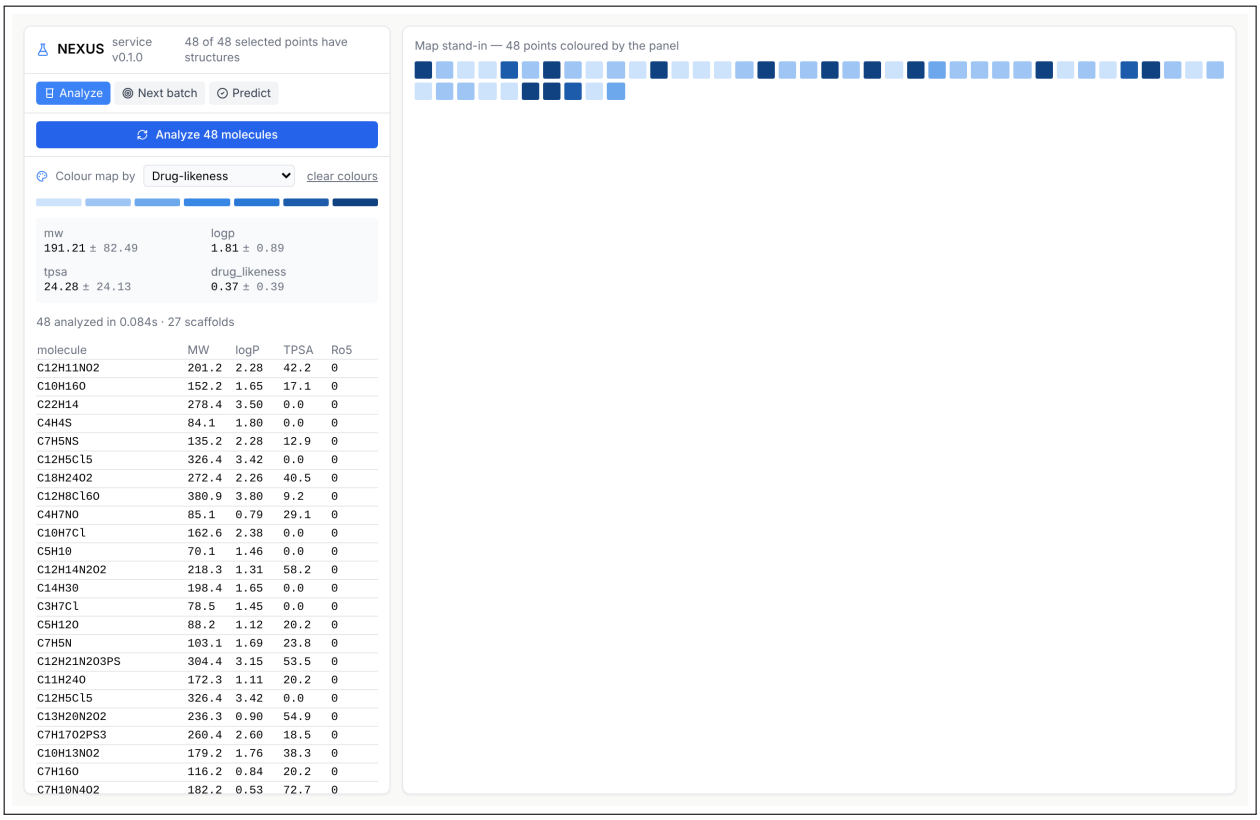


Figure 1: Analyze mode. Forty-eight molecules from the ESOL set, analyzed in 0.084 s into 27 scaffolds. Molecular weight, logP and TPSA are computed by the NumPy package; the right-hand strip is the map, coloured by the chosen descriptor.

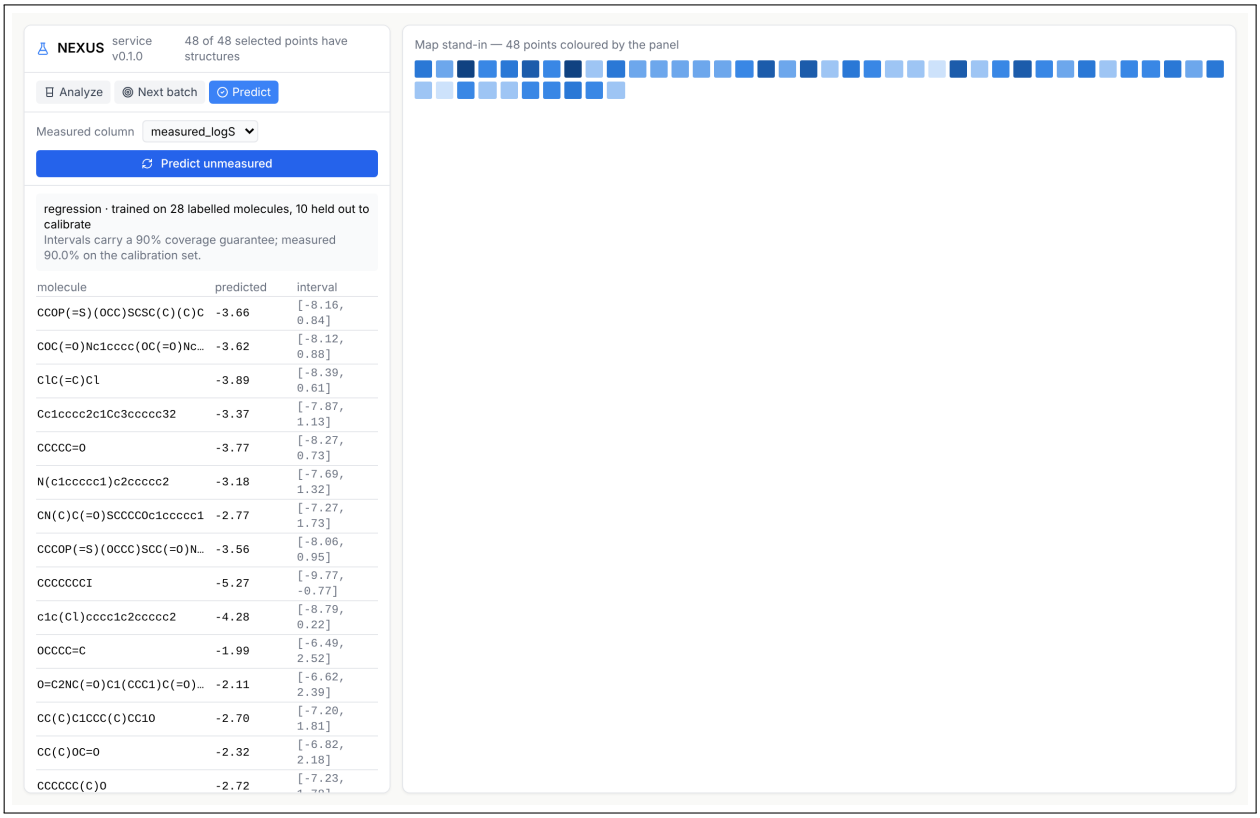


Figure 2: Predict mode, and the result worth singling out. A Tanimoto-kernel Gaussian process is trained on 28 labelled molecules, 10 are held out to calibrate, and the intervals carry a finite-sample 90% coverage guarantee. The panel reports the coverage measured on the calibration set — 90.0% — beside the nominal level.

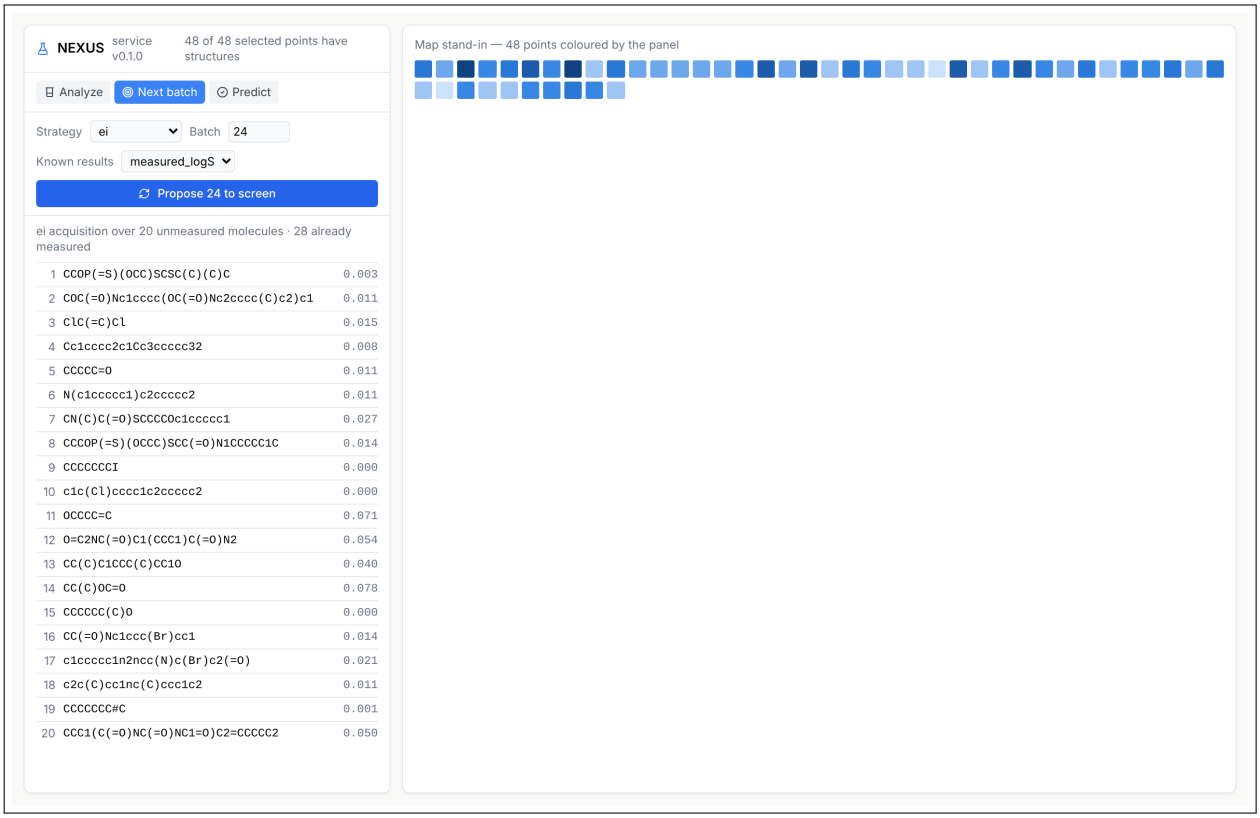


Figure 3: Next-batch mode. The active-learning loop ranks the unmeasured molecules by expected improvement and proposes what to assay next, penalising similarity so one chemical series cannot consume the batch.

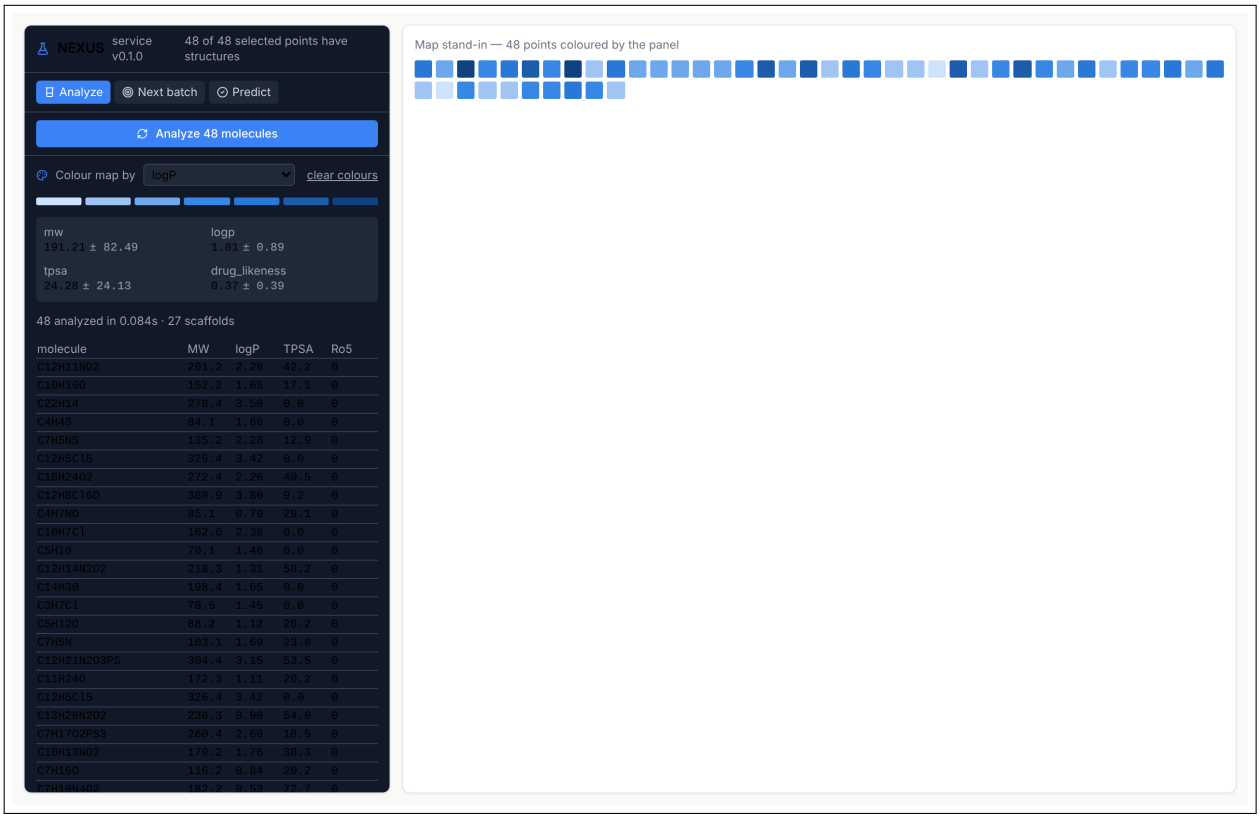


Figure 4: Dark mode, which Mantis panels are expected to support.

baselines beat it on most of these datasets. At 1.5–2k molecules a graph network is not automatically the right tool, and the pipeline reports both rather than only the flattering half.

4 Hit identification

A retrospective screen over 8,000 compounds from the HIV set (3.48% active), screening 13.8% of the library. Each strategy starts from the same random seed batch and may only see labels for compounds it has selected.

strategy	actives recovered	enrichment over random
random	13.9%	1.01× ± 0.10
greedy	42.2%	3.07× ± 0.20
ucb	39.2%	2.85× ± 0.17
ei	41.6%	3.03× ± 0.12
uncertainty	43.4%	3.16× ± 0.23

The random arm landing at 1.01× is the control: it confirms the measurement is unbiased, since random selection should recover actives in proportion to the fraction screened.

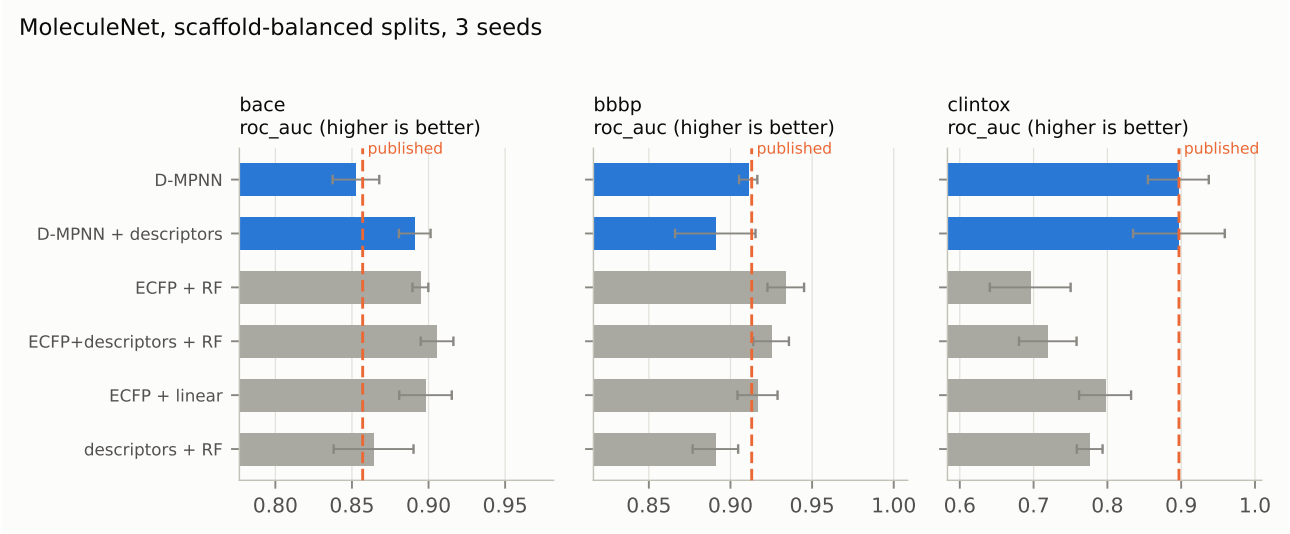


Figure 5: Method comparison per dataset. Blue marks the graph networks, grey the fingerprint and descriptor baselines; the dashed line is the published D-MPNN result. Error bars span three seeds.

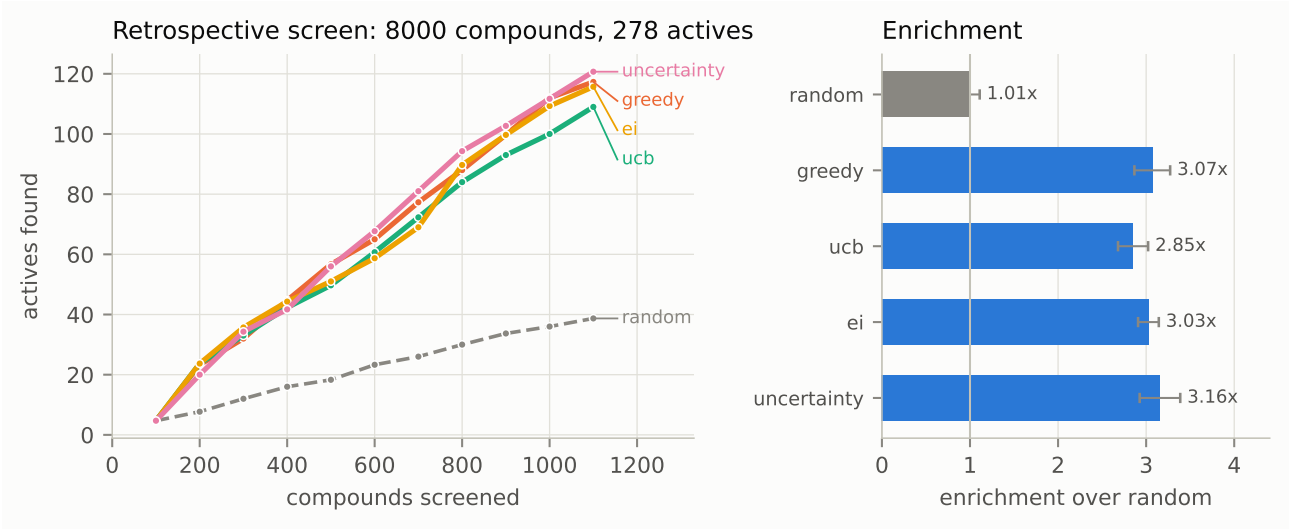


Figure 6: Actives recovered against compounds screened, and the resulting enrichment. Uncertainty sampling and expected improvement both find roughly three times as many actives as random selection at the same budget.

5 Binding affinity

The Davis kinase panel: 68 inhibitors against 442 kinases. Three splits are reported because they answer different questions — random tests interpolation, cold-drug is the lead-discovery case, cold-target the selectivity case.

Davis binding-affinity benchmark still running.

6 Uncertainty that means something

Conformal prediction gives a distribution-free, finite-sample guarantee: if calibration and test points are exchangeable, the region covers the truth with probability at least $1 - \alpha$, whatever the model does. That is a falsifiable claim, so it is tested directly rather than assumed.

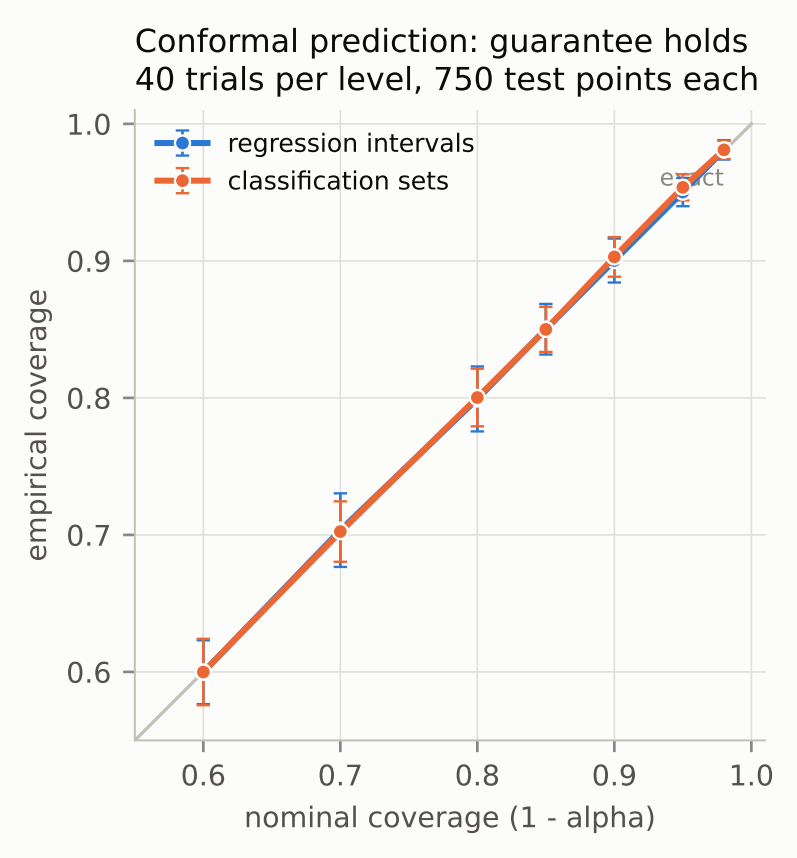


Figure 7: Empirical coverage against the nominal level, over 40 trials at each of seven α values. Both the regression intervals and the classification sets track the diagonal.

7 Free energy

Estimators are validated on a system whose answer is known in closed form: a harmonic oscillator whose spring constant changes from k_0 to k_1, for which $\Delta F = \ln(k_1/k_0)/(2\beta)$ exactly.

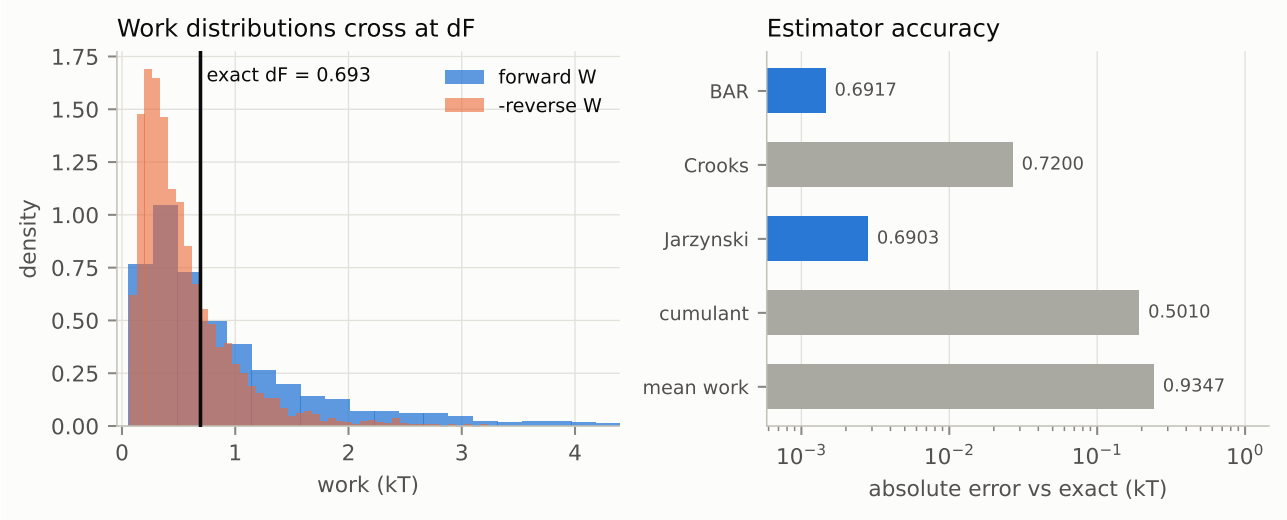


Figure 8: Forward and reverse work distributions cross at ΔF, as Crooks' theorem requires, and the estimator errors against the exact value. BAR is the most accurate, which is what theory predicts; the cumulant expansion is biased, as expected for non-Gaussian work.

8 Descriptors

dataset	reference column	our implementation	n	r	MAE	exact
delaney-processed	Molecular Weight	mw	1128	1.0000	0.071	95.7%
delaney-processed	Number of H-Bond Donors	hbd_strict	1128	0.9981	0.004	99.6%
delaney-processed	Number of Rings	rings	1128	0.9990	0.004	99.6%
delaney-processed	Number of Rotatable Bonds	rotatable_bonds	1128	0.9840	0.230	79.8%
delaney-processed	Polar Surface Area	tpsa	1128	0.9997	0.201	93.4%
bace	MW	mw	1513	1.0000	0.009	100.0%
bace	HBD	hbd_strict	1513	0.9498	0.447	59.7%
bace	HBA	hba_strict	1513	0.8752	0.915	26.0%
bace	RB	rotatable_bonds	1513	0.9993	0.035	96.5%
bace	HeavyAtomCount	heavy_atoms	1513	1.0000	0.000	100.0%
bace	RingCount	rings	1513	1.0000	0.000	100.0%
bace	PSA	tpsa_sp	1513	0.9998	0.114	95.8%

The logP model is the weak one and is reported as such: an atom-contribution fit on 4200 compounds with experimental logD, held-out MAE 0.80 against 0.94 for predicting the mean, R^2 0.21. It is useful as a feature, not as a replacement for a dedicated predictor.

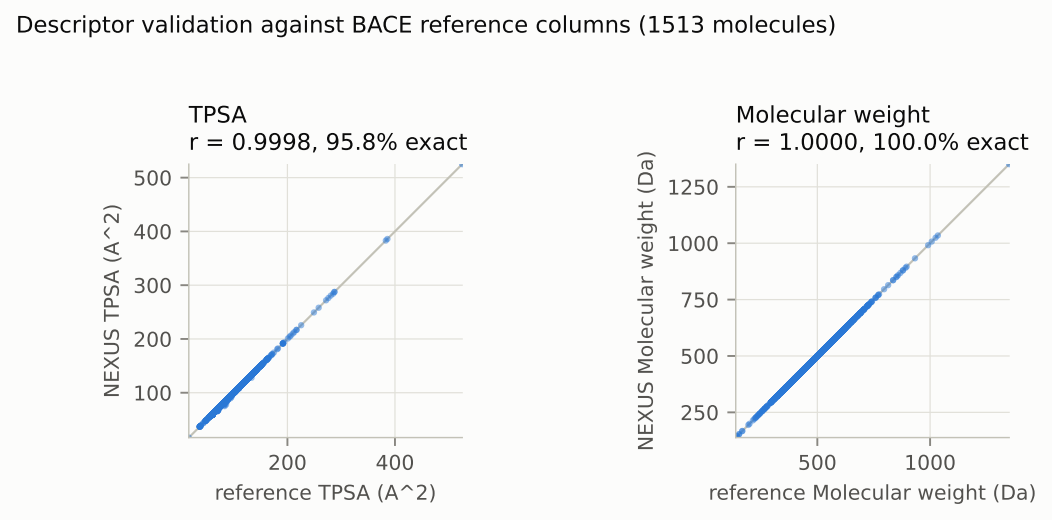


Figure 9: Agreement with the reference descriptor columns shipped in the BACE dataset.

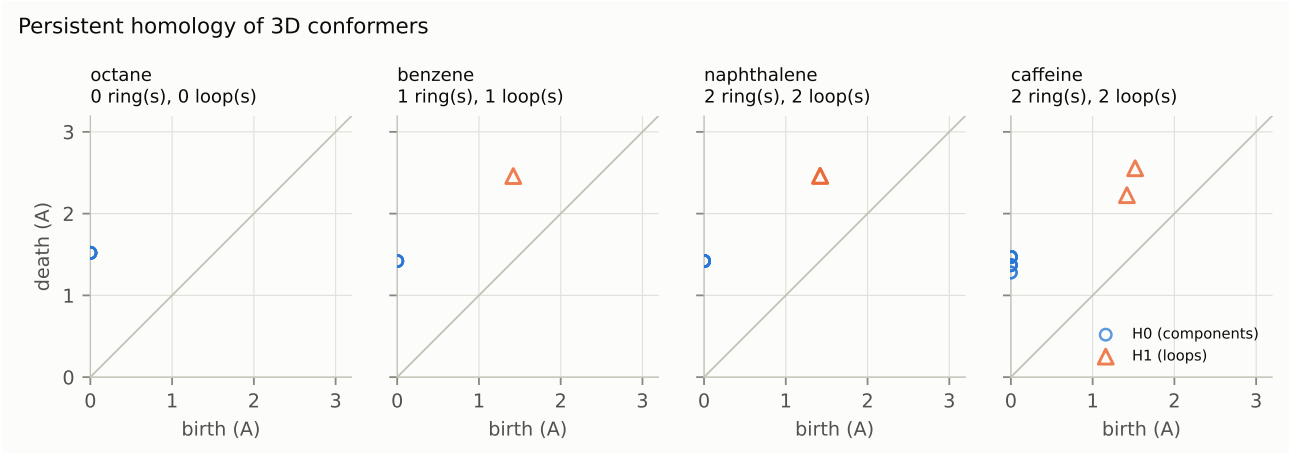


Figure 10: Persistence diagrams of 3D conformers. The number of long-lived H_1 features matches the ring count in each molecule, and acyclic octane produces none.

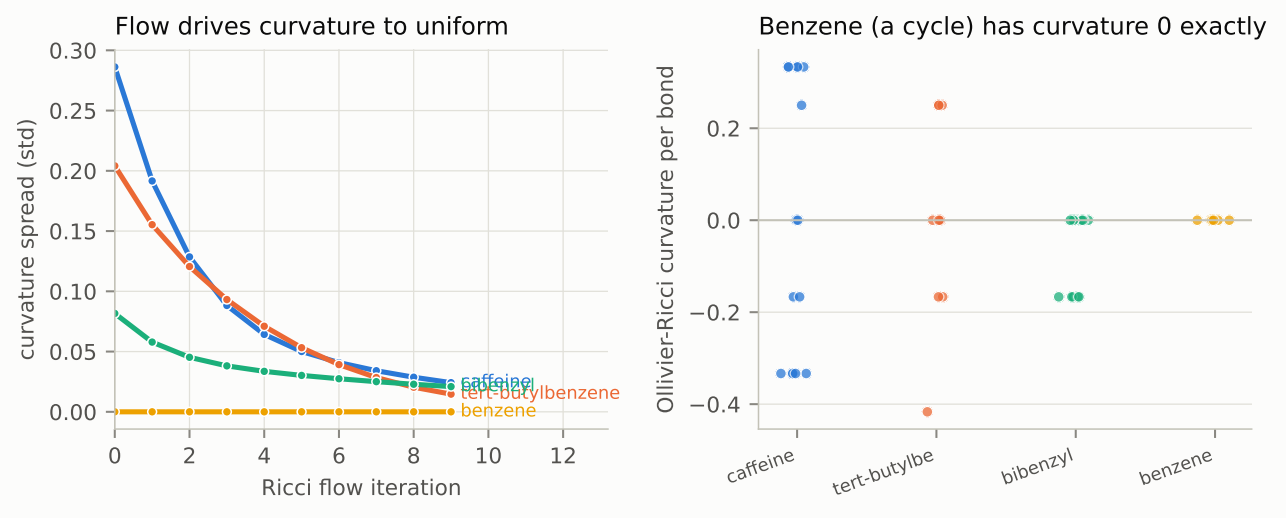


Figure 11: Ricci flow drives the spread of curvature towards zero, and benzene — a cycle graph — has curvature exactly zero, as the theory requires for C_n with $n \geq 6$.

9 Shape and topology

10 Target biology

seeded term	rank recovered	top-ranked term
DNA repair	1	DNA repair
apoptotic process	1	apoptotic process
cell cycle	1	cell cycle
immune response	not recovered	defense response to symbiont
glycolytic process	1	glycolytic process

Random gene sets (40 trials of 60 genes): mean 0.28 significant terms at FDR 0.05, 8% of trials returned any hit. Semantic similarity separates same-pathway from random gene pairs with AUC 0.885 (0.451 versus 0.205).

11 Why scaffold splitting matters

12 De novo generation

sampler	validity	uniqueness	novelty
model + valence constraint	0.0%	0.0%	0.0%
model, unconstrained sampling	0.0%	0.0%	0.0%
marginal baseline (no model)	2.0%	100.0%	100.0%

A negative result, reported. The first generator reached only 3.0% validity, *below* its own 4.0% marginal baseline: sampled bond matrices were far too dense, so valences blew up. Valence-constrained

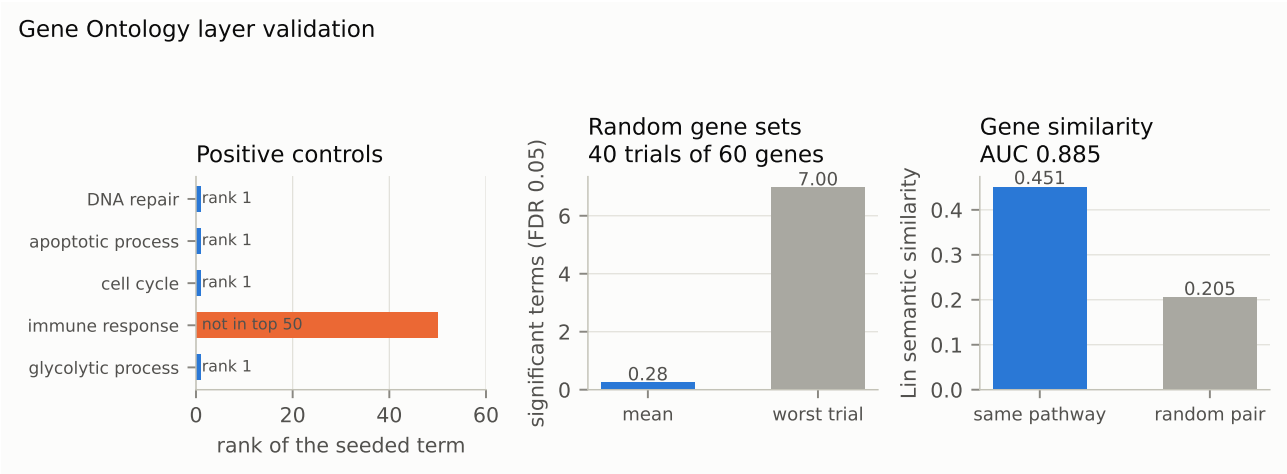


Figure 12: Gene Ontology validation: positive controls recovered, false positives on random gene sets, and semantic similarity separating same-pathway from random gene pairs.

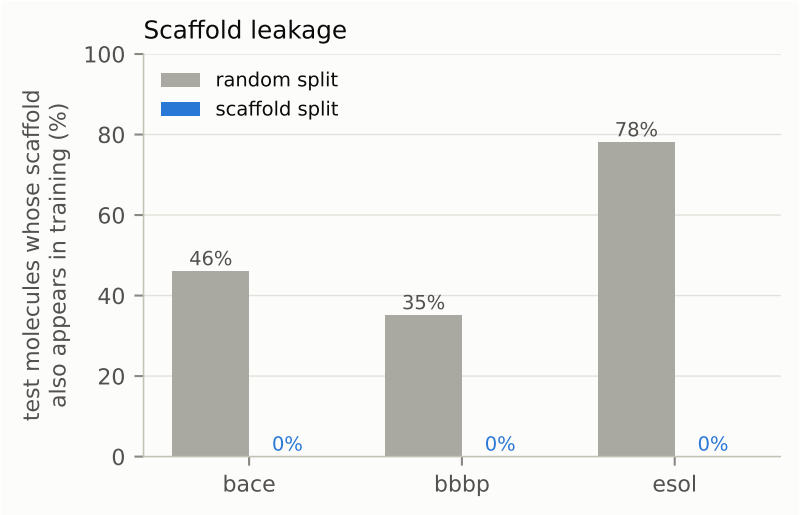


Figure 13: A random split leaves most test molecules sharing a scaffold with the training set, which flatters any model. The scaffold split removes that leakage entirely.

sampling and an eight-fold larger training set were added in response. The table above shows what that actually bought, including the ablation with the constraint switched off.

13 Integration with Mantis

The pipeline ships a drop-in panel for the Mantis map interface. No file in the Mantis repository is modified: three files are copied in and three short registrations are documented as exact diffs. Typechecked against the real Mantis tree with its own `node_modules` and path aliases — 121 pre-existing errors in Mantis's `data_provider.tsx` both with and without the panel, and zero attributable to NEXUS code.

14 Limitations

- Pure NumPy means CPU-bound training. The graph network is smaller and trained for fewer epochs than a GPU implementation would be, so its numbers are a floor rather than a ceiling.
- The logP model is weak (R^2 0.21 on held-out experimental logD).
- The drug-likeness score does *not* separate approved drugs from clinical-trial failures (measured AUC 0.52) and is not a predictor of approval. It is a property-range summary.
- Free energy estimators are validated on analytically solvable systems, not on protein–ligand alchemical transformations, which need a real force field.
- Generation is the weakest component; see the section above.
- Published reference numbers are quoted for orientation. The splits here are regenerated with our own seeds, so they are comparable in protocol but not identical in data.

15 Reproducing everything

```
pip install -e .                                  # NumPy only
python -m pytest -q                               # the test suite
python -m benchmarks.validate_descriptors         # descriptors vs reference values
python -m benchmarks.fit_logp                     # logP fit on experimental logD
python -m benchmarks.validate_go                  # GO enrichment and similarity
python -m benchmarks.validate_conformal           # coverage guarantees
python -m benchmarks.validate_active_learning     # retrospective HIV screen
python -m benchmarks.molnet --seeds 3             # property prediction
python -m benchmarks.affinity                     # Davis binding affinity
python -m benchmarks.generate                     # train and evaluate the generator
python -m benchmarks.figures                      # all figures
python -m benchmarks.report_pdf                   # this document
```